Xinjie Liu

CONTACT INFORMATION

5.104 Peter O'Donnell Jr. Building (POB),
The University of Texas at Austin (UT Austin)
Austin, TX 78712
United States

https://xinjie-liu.github.io
Google Scholar
GitHub
xinjie-liu@utexas.edu

RESEARCH INTERESTS

My research focuses on developing theoretical foundations and practical algorithms for decision-making and control in autonomous systems operating under uncertainty. I am particularly interested in (i) efficient and scalable reinforcement learning through principled integration of data from multiple sources, and (ii) intelligent and safe interaction among multi-agent systems, such as robots operating in shared environments.

EDUCATION

The University of Texas at Austin, Austin, TX
Ph.D. in Electrical and Computer Engineering — Aug 2023 - May 2028 (expected)
Advisors: Ufuk Topcu and David Fridovich-Keil

Delft University of Technology, Delft, Netherlands
M.Sc. in Robotics, GPA: 9.04/10.0 (top 1.6%, *cum laude*) — Sep 2021 - Jul 2023
Advisor: Javier Alonso-Mora

Tongji University, Shanghai, China
B.Eng. in Automotive Engineering, GPA: 88.9/100 — Sep 2016 - Jul 2021

Graz University of Technology, Graz, Austria
Exchange Program — Jan 2020 - Jun 2020

PUBLICATIONS

[11] **X. Liu**, R. Zhao, A. Chaudhuri, C. Neary, U. Topcu, and D. Fridovich-Keil. Multi-Fidelity Policy Gradients Stabilize Data-Scarce Reinforcement Learning, under review, 2026.

[10] H. Xu, P. Srivastava, **X. Liu**, S. Agarwal, and A. Zhang. Horizon-Scalable Q-Learning, under review, 2026.

[9] T. Qiu, F. Fotiadis, **X. Liu**, C. Ellis, J. Milzman, W. A. Suttle, U. Topcu, and D. Fridovich-Keil. Linear-Quadratic Gaussian Games with Distributed Sparse Estimation, IEEE Control Systems Letters (L-CSS), 2026.

[8] **X. Liu***, C. Neary*, K. Gupta, W. A. Suttle, C. Ellis, U. Topcu, and D. Fridovich-Keil. A Multi-Fidelity Control Variate Approach for Policy Gradient Estimation, Transactions on Machine Learning Research (TMLR), 2026. Awarded **J2C Certification** ($<$ 10% of accepted papers); invited to present at NeurIPS 2026.

[7] Y. Jain*, **X. Liu***, L. Peters*, D. Fridovich-Keil, and U. Topcu. Bayesian Inverse Games with High-Dimensional Multi-Modal Observations, under review at The International Journal of Robotics Research (IJRR), 2025.

[6] **X. Liu**, J. Li, F. Fotiadis, M. O. Karabag, J. Milzman, D. Fridovich-Keil, and U. Topcu. Approximate Feedback Nash Equilibria with Sparse Inter-Agent Dependencies, conditionally accepted at IEEE Transactions on Automatic Control (T-AC), 2025.

[5] K. Gupta, **X. Liu**, R. Allen, U. Topcu, and D. Fridovich-Keil. Second-Order Algorithms for Finding Local Nash Equilibria in Zero-Sum Games, under review, 2025.

[4] C. Armstrong*, R. Park*, **X. Liu**, K. Gupta, and D. Fridovich-Keil. Inferring Foresightedness in Dynamic Noncooperative Games, IEEE Robotics and Automation Letters (RA-L), 2025.

*indicates equal contribution.

[3] **X. Liu**, J. Li, F. Fotiadis, M. O. Karabag, J. Milzman, D. Fridovich-Keil, and U. Topcu. Policies with Sparse Inter-Agent Dependencies in Dynamic Games: A Dynamic Programming Approach, Autonomous Agents and Multi-Agent Systems (AAMAS), Extended Abstract, 2025.

[2] **X. Liu***, L. Peters*, J. Alonso-Mora, U. Topcu, and D. Fridovich-Keil. Auto-Encoding Bayesian Inverse Games, Algorithmic Foundations of Robotics XVI (WAFR), 2024.

[1] **X. Liu***, L. Peters*, and J. Alonso-Mora. Learning to Play Trajectory Games against Opponents with Unknown Objectives, IEEE Robotics and Automation Letters (RA-L), 2023.

Technical Reports & Theses

[3] **X. Liu**. On Game-Theoretic Planning with Unknown Opponents' Objectives, master's thesis, Delft University of Technology, 2023. (Mentor: Javier Alonso-Mora)

[2] **X. Liu** and V. Atanassov. Safe Model Predictive Control Approach for Non-Holonomic Mobile Robots, Delft University of Technology, 2022. (Mentor: Sergio Grammatico)

[1] **X. Liu**, R. M. Rodríguez, P. Féry, and Y. Zhang. Planning Algorithm for a Quadrotor Drone, Delft University of Technology, 2022. (Mentor: Javier Alonso-Mora)

Honors and Awards

Cockrell School of Engineering Fellowship, UT Austin 2023-present

J2C Certification, Transactions on Machine Learning Research (< 10% of accepted papers) 2026

Travel Grant, Texas Robotics 2025

Travel Grant, Workshop on the Algorithmic Foundations of Robotics (WAFR) 2024

Graduation *cum laude*, TU Delft (top 1.6% of MSc program) 2023

First Prize, **Franka Emika Robot Hackathon Challenge**, European Robotics Forum 2022

National Scholarship for Undergraduate Exchange Program (1% of department) 2020

Annual Excellent Student Award, Tongji University (top 5% of department) 2020

First Prize Academic Scholarship, Tongji University (top 3% of department) 2019

Winning Prize for Outstanding Innovative Projects, Tongji University 2019
Project: Autonomous Flight of UAVs Based on UWB Localization

Third Prize Academic Scholarship, Tongji University (top 20% of department) 2018

Scholarship for Social Activities, Tongji University 2017, 2019

Projects

Web | GitHub

High-Precision Robot Assembly Challenge Jun 2022 - Jul 2022
Team Member, Delft University of Technology (1/6)
- Developed a point cloud-based perception module and interactive imitation learning technique to tackle high-precision robot assembly tasks on a randomly positioned board
- Winner of the Franka Emika Challenge at the European Robotics Forum Hackathon

Autonomous Robotic Field Coverage Apr 2022 - Jun 2022
Multidisciplinary Project with Lely, Main Contributor (1/5)
- Designed a robotic field coverage system with obstacle avoidance and battery management on a Husky robot, including modules for perception (top-view camera), task planning, path and trajectory planning, and control

MPC Approach for Non-Holonomic Mobile Robots in Dynamic Scenarios Jan 2022 - Apr 2022
Model Predictive Control Course Project, Main Contributor (1/2)
- Proposed an MPC approach with linearized constraints in velocity space for dynamic obstacle avoidance of mobile robots, proved Lyapunov stability for time-varying regulation problems

RNN-Based Event Camera Data Processing Jan 2022 - Apr 2022
Deep Learning Course Project, Main Contributor (1/3)
- Reproduced partial results of "High Speed and High Dynamic Range Video with an Event Camera" using a different dataset, reconstructing intensity images from event data with RNNs

Deep Reinforcement Learning Library Jan 2022 - Apr 2022
Deep Reinforcement Learning Course Project
- Implemented a library including main policy gradient (REINFORCE, Actor-Critic, PPO, DDPG, TD3) and value function (DQN, Double DQN, n-step target, semi-gradients) based methods

Autonomous Quadrotor Delivery System Oct 2021 - Jan 2022
Planning & Decision Making Course Project, Main Contributor (1/4)
- Developed a navigation pipeline for a quadrotor, including global path planning (RRT*), minimum-snap trajectory generation, and nonlinear MPC for tracking and obstacle avoidance

Multisensor Perception for Autonomous Vehicles Oct 2021 - Jan 2022
Machine Perception Course Project
- Developed a self-driving car perception module, including pedestrian detection (CNN, SVM) and vehicle ego-motion compensation using iterative closest point (ICP) with LiDAR data

Learning Autonomous Car Racing from Pixels Sep 2021 - Nov 2021
Machine Learning for Robotics Course Project, Main Contributor (1/2)
- Developed a pipeline for learning driving policies from pixels, including data augmentation, feature extraction, dimensionality reduction, and control (random forest, SVM, neural networks)

Robot Dynamics & Control Sep 2021 - Nov 2021
Robot Dynamics & Control Course Project
- Implemented force and impedance controllers with singularity-robust and task-priority control for a 2-DOF robot arm; implemented PD and nonlinear geometric controllers for quadrotor tracking tasks

Interactive Imitation Learning in Robotics Oct 2020 - Jul 2021
Bachelor's Thesis Project
- Investigated interactive imitation learning methods using human feedback for various simulated robotic tasks, demonstrating improved task performance and data efficiency compared to reinforcement learning baselines

SOFTWARE

MultiFidelityPolicyGradients: Mixes limited target-domain data with abundant low-fidelity data to construct a variance-reduced policy gradient estimator for improved RL sample efficiency

AutoEncodingBayesianInverseGames.jl: Embeds a differentiable Nash game solver into a generative model for amortized inference of continuous distributions in multi-agent interactions

NumericalOptimizationAlgorithms: Numerical optimization algorithms for unconstrained and constrained problems accelerated using JAX

DifferentiableAdaptiveGames.jl: A differentiable Nash equilibrium solver and an inverse game solver for equilibrium-constrained bilevel optimization problems

SafeMPCObstacleAvoidance: MPC planner for non-holonomic robots with linearized dynamic obstacle avoidance constraints in the velocity space

QuadrotorPlanningControl: Navigation pipeline for a quadrotor robot with a global path planner, minimum-snap trajectory generator and MPC controller for obstacle avoidance

MEDIA

Franka Emika Challenge winner at the European Robotics Forum Franka Robotics, 2022

Talks

Texas Symposium on Machine Learning, Responsible AI & Robotics, UT Austin 2026
Boosting Reinforcement Learning with Imperfect Data:
A Multi-Fidelity Policy Gradient Framework

Center for Autonomy Seminar, UT Austin 2025
A Multi-Fidelity Control Variate Approach for Policy Gradient Estimation

Center for Autonomy Seminar, UT Austin 2024
Inferring Multimodal Uncertainty for Game-Theoretic Interactions
Computing Sparse Policies for Noncooperative Dynamic Games

Texas Robotics Seminar, UT Austin 2023
On Game-Theoretic Planning with Unknown Opponents' Objectives

Lockheed Martin Corporation 2023
On Game-Theoretic Planning with Unknown Opponents' Objectives

Princeton University 2023
Learning to Play Trajectory Games Against Opponents with Unknown Objectives

Academic Mentorship

Ryan Park, Computer Science Honors & Aerospace Engineering, UT Austin (next: PhD student at University of California, Berkeley)

Yash Jain, Computational Engineering Honors & Mathematics, UT Austin (next: PhD student at University of Pennsylvania)

Outreach

STEM Muse Mentorship Program, Austin Sep 2024 - present
Mentoring to provide individualized professional development and career support for women

Girlstart Texas, Austin Sep 2023 - present
Organizing innovative workshops to increase girls' interest and engagement in STEM fields

Professional Activities

Reviewer for Journals

- IEEE Transactions on Robotics (T-RO)
- IEEE Transactions on Automatic Control (T-AC)
- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Intelligent Transportation Systems (T-ITS)

Reviewer for Conferences

- Conference on Neural Information Processing Systems (NeurIPS)
- International Conference on Learning Representations (ICLR)
- Reinforcement Learning Conference (RLC)
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- International Symposium on Robotics Research (ISRR)
- IEEE Conference on Decision and Control (CDC)

Graduate-Level Courses

Analysis: Real Analysis, Functional Analysis (basics)
Probability: Probability and Stochastic Processes
Optimization: Convex Optimization, Nonlinear Programming
Machine Learning: Deep Reinforcement Learning, Deep Learning, Machine Learning for Robotics
Control & Game Theory: Game-Theoretic Modeling of Multi-Agent Systems, Model Predictive Control
Robotics: Human-Robot Interaction, Machine Perception, Planning & Decision Making, Robot Dynamics & Control, Robot Software Practicals
Software: Object-Oriented Scientific Programming with C++

SKILLS

Programming Languages: Python, Julia, C++, Matlab
Version Control: Git
Machine Learning & Numerical Computation: PyTorch, JAX, Zygote.jl, Flux.jl, ChainRules.jl, TensorFlow, Keras
Optimization: PATH Solver, JuMP.jl, YALMIP, CasADi, CVX, FORCES Pro, IPOPT, OSQP
Other: Robot Operating System (ROS), Linux, LaTeX
Languages: Mandarin (native), English (C1, IELTS 7.5), German (B2, DSH 2)

MANAGEMENT EXPERIENCE

Business Consulting Intern, IQVIA Shanghai — Jul 2020 - Aug 2020
Marketing project for rare disease drugs: conducted policy and business analysis to develop strategies for national negotiations and local market entry of orphan drugs